

॥ श्रीः ॥ लीलावती

श्रीभूतगणकचक्रचूडामणि—भास्कराचार्यविरचिता

Līlāvatī of Bhāskarācārya
(Treatise on Arithmetic, c. 1150 CE)

Critical edition from the manuscript of the
Late Pt. Manmohan Shastri Collection, Jammu
Digitized by eGangotri

Published by Shyamlal Bhat, Kashi (Varanasi)
Third Edition, Śāka 1829 (1907 CE)

Edited and Typeset in Modern LaTeX
Sanskrit text in Devanagari, English translation, mathematical commentary

Preface

The *Līlāvātī* is the first section of Bhāskara II's monumental work *Siddhāntaśiromaṇi*, composed in 1150 CE. Named after Bhāskara's daughter, it is the most celebrated textbook of Indian arithmetic, covering arithmetic operations with integers and fractions, rules of proportion, interest calculation, plane geometry, solid geometry, and combinatorial problems.

The present edition reproduces the Sanskrit verses (*śloka*) with *anvaya* (word-order analysis) and English translation, together with mathematical explanation. The source is the third edition printed at Kashi (Varanasi) by Shyamlal Bhat, preserved in the collection of the late Pandit Manmohan Shastri of Jammu.

Technical notes: Sanskrit text is set in Noto Serif Devanagari with HarfBuzz rendering. Mathematical formulas follow modern notation. Each rule (*sūtra*) is given in Sanskrit followed by translation and worked examples. All Hindi commentary has been removed; only Sanskrit source text and English translation are retained.

Invocatory Verses (XXXXXXXXXX)

लीलागुल्लक्षोल्कालव्यालविलसिने ।
गणेशाय नमो नीलकमलामलकान्तये ॥ १ ॥

Translation: Salutation to Gaṇeśa, the dark-lotus-hued, the playfully sportive one, having the brightness of the rising sun.

अवतार पुरा पुराणेषां यदुकुले जगतीहितेहिने ।
जयति सोऽयमिमां सकलां लीलावतीं मामपि सिद्धमहीपतिः ॥ ७ ॥

Translation: In former times, the protector of the Yadu dynasty, the benefactor of the world, incarnated. May that same victorious king complete this *Līlāvātī* together with me.

शाकाब्दे वर्षे कविवचनवती त्रिविक्रमो भूतनयोरस्य जातः ।
यो भोजराजेन कृताभिधानो विद्यापतिभूस्करभट्टनामा ॥ ८ ॥

Translation: In the Śaka year, Trivikrama was born, who was given the title Vidyāpati by King Bhoja, known as Bhūskara Bhaṭṭa.

तस्माद्विदुषदर्वज्ञो जातो गोविन्दसन्निभः ।
प्रभाकरः सुतस्तस्यात्मजस्तद्वाऽपरः ॥ ९ ॥

Translation: From him was born the all-knowing scholar Prabhākara, resembling Govinda.

तस्मान्मनोरथो जातः सतां पूर्णमनोरथः ।
श्रीमान्महेश्वराचार्यस्ततोऽजनि कवीश्वरः ॥ १० ॥

Translation: From him was born Manoratha, who fulfilled the desires of the virtuous; and from the illustrious Maheśvara Ācārya was born the lord of poets.

तत्सुतः कविबुधावनिन्दितपदः सदैव विद्यालता-
कण्ठः कविरनुप्रसादितपदः सर्वज्ञविभासदः ।
यच्छिष्यः स ह कोऽपि नो विबुधद्विदक्षो विबाधी क्वचित्
श्रीमान् भास्करकोविदः सममवत्सर्क्षितपुण्यार्णवतः ॥ ११ ॥

Translation: His son, a poet praised by the learned. His student was the illustrious Bhāskara, equally virtuous, ocean of good deeds.

भास्करचितयमथा सिद्धान्तशिरोमणिप्रख्याः ।
तदयं कृताऽनन्ये व्याकृतया मन्मठे नियतं ॥ १५ ॥

Translation: This *Līlāvātī*, composed by Bhāskara, is the jewel on the head of the *Siddhāntaśiromaṇi*, now presented with unique explanations.

Contents

Overview of the Līlāvati

The *Līlāvati* contains the following major topics:

1. ॐॐॐॐॐॐॐॐ (Invocatory Verses) — Salutations and genealogy
2. ॐॐॐॐॐॐ (Definitions) — Units of measurement, numerical notation
3. ॐॐॐॐॐॐॐॐॐॐॐ (Arithmetic) — Operations on whole numbers
4. ॐॐॐॐॐॐॐॐॐ (Fractions) — Operations with fractions
5. ॐॐॐॐॐॐॐॐॐ (Zero) — Operations with zero
6. ॐॐॐॐॐॐॐ (Rule of Three) — Direct and inverse proportion
7. ॐॐॐॐॐॐॐॐ (Mixed Problems) — Interest, alligation, barter
8. ॐॐॐॐॐॐॐॐॐॐॐ (Plane Geometry) — Areas, Pythagorean theorem
9. ॐॐॐॐॐॐ (Excavations, Piles) — Volumes, stacking
10. ॐॐॐॐॐ (Progressions) — Arithmetic and geometric series
11. ॐॐॐॐॐॐॐॐ (Shadows) — Height and distance problems
12. ॐॐॐॐॐ (Pulverizer) — Linear Diophantine equations
13. ॐॐॐॐॐॐ (Permutations) — Combinatorial problems

1 ॐॐॐॐॐॐ — Definitions and Units of Measurement

The *Līlāvati* begins with fundamental definitions of units of measurement essential for all practical arithmetic in medieval India.

1.1 ॐॐ ॐॐॐॐॐॐ — Units of Weight

पादोनग्रानकतुल्यष्टकैः समतुल्यैः कथितोऽत्र सेरः ॥
मणाभिधानं व्युगमैः सैरेधार्यादितोल्येषु तुष्टकसङ्ख्या ॥ १ ॥

Translation: A *ser* is equal to eight *pāda* or equal to eight grains. A *maṇa* is twice the amount of a *ser* in weighing gold, etc.

द्वयकद्विसहस्रैर्घटकैः सैरस्तैः पञ्चभिः स्याद्धटिका च ताभिः ॥
मणोऽष्टमित्वालिमगीरशाहकृतास्त्र सङ्ख्या निजराज्यपृष्ठे ॥ २ ॥

Translation: Two thousand *ghaṭakas* make a *ser*; five of these *sers* make a *ghaṭikā*. The weight of one *maṇa* in this kingdom follows this system.

1.2 ॐॐॐॐॐॐॐॐ — Definitions of Time

शोभाः कालादिपरिभाषा लोकतः प्रसिद्धा देया ॥

Translation: The definitions of time and other measures are well-known in the world.

The traditional Indian system of time measurement: 60 *vipalas* = 1 *pala*; 60 *palas* = 1 *ghaṭikā* (24 minutes); 60 *ghaṭikās* = 1 day and night; 15 days = 1 *pakṣa* (fortnight); 2 *pakṣas* = 1 month; 12 months = 1 year.

2 ॐॐॐॐॐॐॐॐॐ — Arithmetic Operations

This chapter covers the fundamental operations: addition, subtraction, multiplication, division, squaring, square roots, cubing, and cube roots.

2.1 ॐॐॐॐॐॐॐॐॐ — Addition and Subtraction

सूत्रः यथोक्तं परिकल्प्य गणितं कुर्यात्।

As taught, having set up the numbers, one should compute.

Numbers must be aligned by place value before performing addition or subtraction. The commentary demonstrates this with worked examples.

Example: Add 2, 5, 32, 193, 18, 10, and 100.

Set up and sum: $2 + 5 + 32 + 193 + 18 + 10 + 100 = 360$.

Example: Subtract 10000 from 360: $10000 - 360 = 9640$.

Example: $\sqrt[3]{1728} = 12$, since $12^3 = 1728$.

1. (*bhinna*) – Simple fraction $\frac{a}{b}$
2. (*prabhāga*) – Part of a part
3. (*bhāgānubandha*) – Fraction combined with an integer
4. (*bhāgāpavāha*) – Fraction less than an integer

.....

$$\text{karna} = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15$$

8.2 ॐ — Piles of Bricks

This section deals with stacking problems and the calculation of volumes of piles.

Problem: Bricks are piled in the form of a pyramid with a rectangular base. The base has 10 bricks along one side and 6 along the other. The pile is 5 layers high. Find the total number of bricks.

$$N = 10 \cdot 6 + 9 \cdot 5 + 8 \cdot 4 + 7 \cdot 3 + 6 \cdot 2 = 60 + 45 + 32 + 21 + 12 = 170$$

For square pyramids, the formula is:

$$N = \frac{n(n+1)(2n+1)}{6}$$

8.3 ॐ — Progressions

श्रेण्यांसा ---

The *śreṇī* (progression) section covers arithmetic and geometric progressions.

ॐ — Arithmetic Progression

सूत्र: आदिरुत्तरगुणितः पूर्विका सङ्कलितो भवति ॥

The first term multiplied by the common difference, combined with the sum of the preceding terms...

For an arithmetic progression with first term a , common difference d , and n terms:

- n -th term: $a_n = a + (n - 1)d$
- Sum of n terms: $S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$

where l is the last term.

Example: Find the sum of the first 100 natural numbers.

$$S_{100} = \frac{100}{2}(1 + 100) = 50 \times 101 = 5050$$

ॐ — Geometric Progression

सूत्र: गुणोत्तरे गुणकसङ्गुणिते पूर्वसमासः ॥

In a geometric progression, multiply the first term by the ratio raised to the power of the number of terms...

For a geometric progression with first term a and common ratio r :

- n -th term: $a_n = a \cdot r^{n-1}$
- Sum of n terms: $S_n = a \cdot \frac{r^n - 1}{r - 1}$

Problem: A king offers a reward of 1 grain of rice on the first square of a chessboard, 2 on the second, 4 on the third, and so on, doubling each time. Find the total grains on all 64 squares.

$$S_{64} = 1 + 2 + 4 + \dots + 2^{63} = \frac{2^{64} - 1}{2 - 1} = 2^{64} - 1 = 18446744073709551615$$

9 ॐ — Shadow Problems

छायाव्यवहारकथन ---

This chapter uses the properties of similar triangles to solve problems involving shadows, heights of objects, and distances.

9.1 ञायाग्रदूरं द्विजोत्कर्षं कुर्यात् — Finding the Height of a Lamp

सूत्रः छायाग्रदूरं द्विजोत्कर्षं कुर्यात्।

The distance to the tip of the shadow multiplied by the height of the object, divided by the length of the shadow, gives the height of the light source.

Problem: A lamp is placed on a pillar. A man of height 6 stands 8 units away from the pillar, and his shadow extends 12 units from his feet. Find the height of the lamp.

Solution: Using similar triangles:

$$\frac{h}{6} = \frac{8 + 12}{12} = \frac{20}{12}$$

$$h = 6 \times \frac{20}{12} = 10 \text{ units}$$

9.2 ञायाग्रदूरं द्विजोत्कर्षं कुर्यात् — Difference of Two Shadows

When an object casts different shadows from two different light sources:

$$\frac{h_1}{s_1} = \frac{h_2}{s_2} = \frac{d}{s_1 - s_2}$$

10 कुट्टकव्यवहारः — The Pulverizer

कुट्टकव्यवहारः ---

The *kuṭṭaka* (pulverizer) is the method for solving linear Diophantine equations of the form:

$$ax + by = c$$

where a , b , and c are given integers, and x , y are unknown integers to be found.

10.1 कुट्टकव्यवहारः — Fixed Pulverizer

सूत्रः द्विच्छेदः कुट्टकः---

The pulverizer is the division of two (numbers).

The method uses the Euclidean algorithm to find the greatest common divisor, then back-substitutes to find the solution.

Problem: Find x and y such that $121x + 17 = y$

Solution: Using the *kuṭṭaka* method:

Step 1: Apply the pulverizer to 121 and 17:

$$121 = 7 \times 17 + 2$$

$$17 = 8 \times 2 + 1$$

Step 2: Work backwards to express 1 as a combination of 121 and 17:

$$1 = 17 - 8 \times 2$$

$$= 17 - 8 \times (121 - 7 \times 17)$$

$$= 17 - 8 \times 121 + 56 \times 17$$

$$= 57 \times 17 - 8 \times 121$$

So $x = 8$, $y = 121 \times 8 + 17 = 985$ gives a solution.

10.2 ञायाग्रदूरं द्विजोत्कर्षं कुर्यात् — Transposition Method

The transposition method converts a given equation into the standard form by rearrangement.

$$h^2 + 2ha + a^2 = h^2 + d^2$$

$$h = \frac{d^2 - a^2}{2a}$$

Example: If the lotus stands 1 *āṅgula* above the water and is blown 2 *hastas* sideways:

$$h = \frac{2^2 - (1/24)^2}{2 \times (1/24)} \approx \frac{4}{1/12} = 48 \text{ āṅgulas}$$

The Pearl Necklace Problem (XXXXXXXXXXXXXX)

मुक्ताहारः स्तनान्तःस्थितो मणिः श्रेण्यां पतन् वेगात् ।
स्तनभूमिं प्राप्य त्रिभागं श्रेण्याः प्रचक्रमे पद्मे ॥

Translation: A string of pearls with a jewel at the center was hanging down. The necklace broke and the pearls scattered. One-third of the pearls fell to the ground, one-fifth remained on the couch, one-sixth was saved by the lady, one-tenth was caught by her lover, and six pearls remained on the string. Tell me the total number of pearls in the necklace.

Solution: Let x be the total number of pearls.

The fraction accounted for is:

$$\frac{1}{3} + \frac{1}{5} + \frac{1}{6} + \frac{1}{10} = \frac{10 + 6 + 5 + 3}{30} = \frac{24}{30} = \frac{4}{5}$$

The remaining pearls = $x - \frac{4x}{5} = \frac{x}{5} = 6$, so $x = 30$.

Answer: 30 pearls.

The Peacock and the Snake (XXXXXXXXXXXXXX)

मयूरोऽङ्गनकूटे स्थितो बिलात्सर्पो दूरतः पश्यन् ।
पतत्यरेखां कृत्वा यावत् सर्पो विलं गतस्तावत् ॥

Translation: A peacock is perched on the top of a pillar. A snake is at the base of the pillar, at some distance from it. The peacock pounces on the snake in a straight line, covering equal distances down the pillar and along the ground. Find the point where they meet.

Mathematical formulation: Let the pillar height be P and the snake's distance from the base be D . The peacock flies in a straight line to meet the snake. If the meeting point is at distance x from the base:

$$P - x = \sqrt{x^2 + D^2}$$

Squaring both sides:

$$P^2 - 2Px + x^2 = x^2 + D^2$$

$$x = \frac{P^2 - D^2}{2P}$$

Example: Pillar height = 9, snake's distance = 27:

$$x = \frac{81 - 729}{18} = \frac{-648}{18} = -36$$

This negative value indicates the formulation; the actual problem as stated in the *Līlāvātī* has the peacock flying along the hypotenuse.

The Swallow Problem (XXXXXXXXXX)

Problem: Swallows are sitting on the branches of a tree. If each branch has 3 swallows, one swallow is left over. If each branch has 4 swallows, one branch is left empty. How many swallows and branches are there?

Solution: Let s = number of swallows, b = number of branches.

$$s = 3b + 1$$

$$s = 4(b - 1) = 4b - 4$$

Equating: $3b + 1 = 4b - 4$, so $b = 5$ and $s = 16$.

Answer: 16 swallows and 5 branches.

The Elephant and the Shepherd (XXXXXXXXXXXXXXXXXX)

गजानां षोडशदन्तैर्मिथोऽभिमुखयोर्द्वयोः संयुक्तयोः ।
दन्तानां सङ्ख्यां वद त्वं यदि गणिते कुशलोऽसि ॥

Translation: Two elephants, each having 16 tusks (8 on each side), face each other in battle. If their tusks become interlocked, tell me the total number of tusks.

Solution: A simple counting problem: each elephant has 16 tusks, so together they have $16 + 16 = 32$ tusks. The problem tests the student's ability to extract relevant information from a poetic description.

The Snake and the Peacock (XXXXXXXXXXXXXXXXXX)

सर्पोऽप्यङ्गनकूटे मयूरं दृष्ट्वा तरसा पतति ।
यावदन्तर्भूमौ गच्छति तावत्पतत्यपि स मयूरः ॥

Translation: A peacock is sitting on a pillar of height 9 *hastas*. A snake is at the base of the pillar, at a distance of 27 *hastas*. Seeing the snake, the peacock pounces in a straight line. Meanwhile, the snake also moves toward the pillar. Find the distance from the pillar where they meet.

Solution: This is a classic pursuit problem. If the peacock flies along the hypotenuse and the snake crawls along the ground, meeting at distance x from the pillar:

$$\text{Peacock's path} = \sqrt{9^2 + (27 - x)^2}$$

Assuming equal speeds, they meet where the peacock's flight distance equals the snake's remaining distance. Setting up the equation and solving gives the meeting point.

Appendix: Sanskrit Mathematical Vocabulary

Numbers and Operations

	Sanskrit	Transliteration	Meaning
सङ्ख्या	<i>saṅkhyā</i>	number	
अङ्क	<i>aṅka</i>	digit	
शून्य	<i>śūnya</i>	zero	
एक	<i>eka</i>	one	
दश	<i>daśa</i>	ten	
शत	<i>śata</i>	hundred	
सहस्र	<i>sahasra</i>	thousand	
कोटि	<i>koṭi</i>	ten million	
योग	<i>yoga</i>	addition	
व्यवकलन	<i>vyavakalana</i>	subtraction	
गुणन	<i>guṇana</i>	multiplication	
भागहार	<i>bhāgaḥāra</i>	division	
वर्ग	<i>varga</i>	square	
वर्गमूल	<i>vargamūla</i>	square root	
घन	<i>ghana</i>	cube	
घनमूल	<i>ghanamūla</i>	cube root	
भिन्न	<i>bhinna</i>	fraction	
रूप	<i>rūpa</i>	integer (unit)	
प्रमाण	<i>pramāṇa</i>	argument (in rule of three)	
फल	<i>phala</i>	result	
इच्छा	<i>icchā</i>	requisition	

Geometric Terms

Sanskrit	Transliteration	Meaning
क्षेत्र	<i>kṣetra</i>	plane figure, field
भुजा	<i>bhujā</i>	side, base
कोटि	<i>koṭi</i>	perpendicular
कर्ण	<i>karṇa</i>	hypotenuse, diagonal
व्यास	<i>vyāsa</i>	diameter
व्यासार्ध	<i>vyāsārdha</i>	radius
परिधि	<i>paridhi</i>	circumference
दैर्घ्य	<i>dairghya</i>	length
विस्तार	<i>vistāra</i>	breadth
ऊन	<i>ūna</i>	less, diminished
अधिक	<i>adhika</i>	more, increased
क्षेत्रफल	<i>kṣetraphala</i>	area
घनफल	<i>ghanaphala</i>	volume

Colophon

Līlāvātī of Bhāskarācārya

Critical edition prepared from the manuscript preserved in the

Late Pt. Manmohan Shastri Collection, Jammu

Digitized by eGangotri

Original printed edition:

Published by **Shyamlal Bhat**, Kashi (Varanasi)

Third Edition, Śāka 1829 (1907 CE)

Typeset in L^AT_EX using:

XeLaTeX with Noto Serif Devanagari

and Linux Libertine O fonts

“Līlāvātī has been the most beloved of all mathematical textbooks in India.

*Named after Bhāskara’s daughter, it has instructed generations
of students in the beautiful art of calculation.”*